

1.1 Použitím vlastností skalárního součinu dostáváme:

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 + \langle u, v \rangle + \langle u, v \rangle + \|v\|^2$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 - \langle u, v \rangle - \langle u, v \rangle + \|v\|^2$$

$$\|u+v\|^2 - \|u-v\|^2 = \|u\|^2 - \langle u, v \rangle + \langle u, v \rangle + \|v\|^2 - \|u\|^2 + \langle u, v \rangle + \langle u, v \rangle - \|v\|^2 = 2\langle u, v \rangle + 2\langle u, v \rangle$$

$$\text{Poznámka: } z + \bar{z} = 2\operatorname{Re}(z), \text{ ale } 4\operatorname{Re}(u, v) = \|u+v\|^2 - \|u-v\|^2, \text{ ale } \operatorname{Re}(u, v) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle u, v \rangle - \langle v, v \rangle = \|u\|^2 + \langle u, v \rangle + \langle u, v \rangle - \|v\|^2$$

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle u, v \rangle + \langle v, v \rangle = \|u\|^2 + \langle u, v \rangle + \langle u, v \rangle + \|v\|^2$$

$$\|u-v\|^2 - \|u+v\|^2 = \|u\|^2 + \langle u, v \rangle + \langle u, v \rangle - \|v\|^2 - \|u\|^2 - \langle u, v \rangle - \langle u, v \rangle - \|v\|^2 = -4\operatorname{Im}(u, v)$$

$$\text{Poznámka: } \|u-v\|^2 - \|u+v\|^2 = -4\operatorname{Im}(u, v), \text{ ale } \operatorname{Im}(u, v) = \frac{1}{2i}(\|u-v\|^2 - \|u+v\|^2)$$

$$\text{Poznámka: } \bar{z} + z = 2\operatorname{Re}(z), \text{ ale } -\bar{z} + z = 2i\operatorname{Im}(z)$$

$$\hookrightarrow i(\bar{z} - z) = 2\operatorname{Im}(z) \hookrightarrow i(\bar{z} - z) = 2\operatorname{Im}(z)$$